

MPUMALANGA MINING SOLUTIONS (MMS)

EXECUTIVE MASTER BLUEPRINT & PRODUCT REQUIREMENTS DOCUMENT (PRD)

Version 1.0

Confidential Internal Document

EXECUTIVE SUMMARY

Mpumalanga Mining Solutions (MMS) is a digital-first mining and machinery training institution designed to operate efficiently with minimal physical infrastructure while delivering a professional, scalable, and technology-driven student experience.

The platform serves as:

Training Institution

Digital Campus

Student Management System

Learning Management System

Marketing Platform

Admissions Office

Communication Hub

Certificate Management System

AI-Powered Operations Center

The long-term vision is to establish MMS as one of South Africa's leading machinery and industrial training institutions.

BUSINESS OVERVIEW

Business Name:

Mpumalanga Mining Solutions

Abbreviation:

MMS

Location:

Middelburg, Mpumalanga

Industry:

Mining Training

Machinery Training

Industrial Skills Development

Safety Training

Business Model:

Training Institution

Digital Learning Platform

Student Services

Certification Facilitation

Corporate Training

Future Recruitment Services

VISION

To become South Africa's most trusted digital-first mining and machinery training institution.

MISSION

To provide practical, industry-focused training that prepares students for opportunities within mining, construction, engineering, and industrial sectors.

CORE VALUES

Professionalism

Safety

Integrity

Innovation

Excellence

Accountability

Student Success

Continuous Improvement

BUSINESS OBJECTIVES

Generate Student Enrollments

Deliver Quality Training

Automate Operations

Reduce Administrative Costs

Increase Accessibility

Support Student Success

Scale Nationally

Establish Industry Trust

PRIMARY PROBLEM BEING SOLVED

Many training institutions depend heavily on physical offices and manual processes.

MMS solves this through:

Digital Enrollment

Automated Communication

Online Learning

AI Assistance

Centralized Administration

Mobile Accessibility

TARGET AUDIENCE

School Leavers

Unemployed Youth

TVET Graduates

Mining Workers

Construction Workers

Industrial Workers

Machine Operators

Corporate Clients

Employers

Government Programs

PRODUCT ECOSYSTEM

The MMS platform consists of:

Public Website

Enrollment System

Student Portal

Learning Platform

Admin Dashboard

AI Assistant

Communication System

Marketing System

Reporting System

Certificate Verification System

TECHNOLOGY STACK

Frontend:

Next.js

React

TypeScript

TailwindCSS

Shadcn UI

Framer Motion

Backend:

Next.js Server Actions

Supabase

PostgreSQL

Supabase Auth

Supabase Storage

Communications:

Resend

Email Automation

Portal Notifications

Future WhatsApp Integration

Artificial Intelligence:

NVIDIA APIs

AI Receptionist

AI Admissions Assistant

AI Learning Assistant

AI Support Assistant

AI Administrative Assistant

Hosting:

Vercel

Analytics:

Google Analytics

Google Search Console

Microsoft Clarity

USER TYPES

Visitor

Lead

Applicant

Student

Instructor

Administrator

Super Administrator

Future Employer

Future Partner

DIGITAL ENROLLMENT SYSTEM

Capabilities:

Online Applications

Manual Form Downloads

Document Uploads

Application Tracking

Portal Account Creation

Email Automation

Instructor Notifications

STUDENT PORTAL

Capabilities:

Course Access

Learning Materials

Progress Tracking

Notifications

Certificates

Invoices

Receipts

Support Requests

Profile Management

Document Center

LEARNING MANAGEMENT SYSTEM

Capabilities:

Modules

Lessons

Videos

Assessments

Downloads

Progress Tracking

Completion Tracking

Certificates

Instructor Support

Learning Analytics

ADMIN DASHBOARD

Capabilities:

Application Management

Student Management

Course Management

Content Management

Payment Verification

Certificate Management

Marketing Management

Reporting

AI Monitoring

Support Management

AI WORKFORCE

MMS Assistant

Admissions Agent

Learning Assistant

Support Agent

Marketing Assistant

Administrative Assistant

Document Assistant

AUTOMATION LAYER

Automatic Emails

Automatic Notifications

Application Routing

Student Onboarding

Document Generation

Certificate Generation

Follow-Up Tasks

Lead Nurturing

COMMUNICATION SYSTEM

Application Emails

Approval Emails

Payment Emails

Course Notifications

Portal Notifications

Newsletters

Marketing Campaigns

Support Responses

MARKETING SYSTEM

SEO

Content Marketing

Social Media

Email Marketing

Lead Magnets

Google Business Profile

Course Landing Pages

Analytics

Conversion Tracking

COURSES

Forklift

Excavator

Front End Loader

TLB

ADT

Dump Truck 777D

Tracked Dozer

Grader

Roller

Tractor

Bobcat

LHD Scoop

Drill Rig

Mobile Crane

First Aid

Fire Fighting

SHE Representative

SECURITY FRAMEWORK

POPIA Compliance

Role-Based Access

Row Level Security

Encrypted Storage

Audit Logs

Certificate Verification

Backup Procedures

Incident Management

DESIGN SYSTEM

Industrial Black

Gold

Silver

White

Premium Industrial Branding

Mobile First

Accessibility Compliant

High Conversion Design

Professional User Experience

SEO STRATEGY

Mining Training Mpumalanga

Excavator Training

Forklift Training

Machine Operator Training

Heavy Machinery Training

Mining Courses South Africa

Local SEO

Course SEO

Content Marketing

Blog System

REPORTING SYSTEM

Applications

Students

Payments

Revenue

Certificates

Marketing

Support

AI Performance

Instructor Performance

SUCCESS METRICS

Lead Generation

Applications

Enrollments

Revenue

Course Completion

Student Satisfaction

Support Resolution

SEO Rankings

Conversion Rates

Retention Rates

DEVELOPMENT PHASES

Phase 1

Public Website

Enrollment

Email Automation

SEO Foundation

Phase 2

Admin Dashboard

Student Management

Payments

Operations

Phase 3

Student Portal

Certificates

Document Center

Phase 4

Learning Management System

Assessments

Progress Tracking

Instructor Tools

Phase 5

AI Workforce

Automation

Document Generation

Advanced Support

Phase 6

Marketing Automation

Lead Nurturing

Advanced Analytics

Growth Systems

Phase 7

National Expansion

Employer Portal

Corporate Training

Recruitment Services

Multi-Campus Support

LONG-TERM VISION

MMS evolves from:

Training Institution



Digital Training Institution



National Skills Development Platform



Industry Workforce Development Platform



MASTER PRINCIPLE

Every feature, page, workflow, automation, email, dashboard, report, and AI capability must answer one question:

"Does this improve student success, business growth, operational efficiency, or trust?"

If the answer is no, the feature should not be built.

FINAL OBJECTIVE

Create a modern, scalable, technology-driven training institution capable of operating efficiently without a traditional office while delivering a world-class student experience.

Mpumalanga Mining Solutions must become the benchmark for digital-first machinery training institutions in South Africa.

This document serves as the master blueprint governing all future design, development, operations, marketing, AI systems, learning systems, and business growth initiatives.